Published in the Journal of AIS

Knowledge Management Perspective of Generative Artificial Intelligence (GenAI)

Maryam Alavi[1], Dorothy E. Leidner[2], Reza Mousavi[2]

[1] Scheller College of Business, Georgia Institute of Technology
[2] McIntire School of Commerce, University of Virginia

Abstract

In this editorial, revisiting Alavi and Leidner (2001) as a conceptual lens, we consider the organizational implications of generative artificial intelligence (GenAI) from a knowledge management (KM) perspective. We examine how GenAI impacts the processes of knowledge creation, storage, transfer, and application, highlighting both the opportunities and challenges this technology presents. In knowledge creation, GenAI enhances information processing and cognitive functions, fostering individual and organizational learning. However, it also introduces risks like AI bias and reduced human socialization, potentially marginalizing junior knowledge workers. For knowledge storage and retrieval, GenAI's ability to quickly access vast knowledge bases significantly changes employee interactions with KM systems. This raises questions about balancing human-derived tacit knowledge with AI-generated explicit knowledge. The paper also explores GenAI's role in knowledge transfer, particularly in training and cultivating a learning culture. Challenges include an overreliance on AI and risks in disseminating sensitive information. In terms of knowledge application, GenAI is seen as a tool to boost productivity and innovation, but issues like knowledge misapplication, intellectual property, and ethical considerations are critical. Conclusively, the paper argues for a balanced approach to integrating GenAI into KM processes. It advocates for harmonizing GenAI's capabilities with human insights to effectively manage knowledge in contemporary organizations, ensuring both technological advances and ethical responsibility.

Keywords: Generative AI (GenAI), knowledge management (KM), large language models (LLMs), knowledge workers

Recommended Citation: Alavi, Maryam; Leidner, Dorothy E.; and Mousavi, Reza (2024) "A Knowledge Management Perspective of Generative Artificial Intelligence," Journal of the Association for Information Systems, 25(1), 1-12. DOI: 10.17705/1jais.00859. Available at: https://aisel.aisnet.org/jais/vol25/iss1/15

Electronic copy available at: https://ssrn.com/abstract=4782875

Introduction

In the mid-1990s, the knowledge-based view of the firm arose as a new way of thinking about organizations. The knowledge-based view focuses on understanding organizations as knowledge production and application systems. Instead of viewing organizations as producing products and services with processes in place to facilitate production, organizations came to be regarded as orchestrating knowledge resources and processes to create and apply knowledge, with products and services outcome of the knowledge production processes. Knowledge thus became a central focus of organizational strategizing with the capacity of firms to generate, share, and leverage knowledge -- particularly propriety knowledge – recognized as paramount to a firm's ability to compete and offer innovative products and services. Knowledge management initiatives in the early 2000s were strongly promoted, with much hype surrounding their value-adding potential and their success stories. Although knowledge management has, for the most part, faded from organizational parlance, the notion of knowledge as a foundational organizational resource and organizations as creators, purveyors, and appliers of knowledge remains intact. Against this understanding that knowledge is a foundational element of organizational success, we pause to reflect on the implications of a new technology – GenAI -- for knowledge and knowledge management in organizations.

Recent advances in AI, especially the emergence of powerful tools called generative AI (GenAI), have launched a new, exciting, and complex era of organizational knowledge management.[1] Not surprisingly, large firms in consulting, banking, and financial services are investing heavily in

[1] Although GenAI is referred to a variety of generative AI models that can produce different forms of artifacts such as text (e.g., ChatGPT by OepnAI), image (e.g., DALL-E by OpenAI), sound (e.g., AudioCraft by Meta), and video (e.g., Gen2 by Runway AI), our primary emphasis is on the GenAI models dedicated to text generation (i.e., large language models- LLMs), given their significant relevance to knowledge management activities within organizations. For a comprehensive list of GenAI applications for generating various types of artifacts, see https://research.aimultiple.com/generative-ai-applications/.

Electronic copy available at: https://ssrn.com/abstract=4782875

GenAI to support their knowledge workers and enhance their products, services, and customer support. For example, McKinsey & Company (2023) recently announced the launch of Lilli, a GenAI-based knowledge management system (KMS) designed to facilitate streamlined search, access, analysis, and synthesis of the firm's knowledge resources for its workforce. The system also aids in identifying domain experts within the firm. Similarly, Morgan Stanley has developed and implemented a KMS to assist its financial advisors (Son, 2023). The system is based on OpenAI's GenAI software, GPT-4, and draws upon a large set of internal documents, market research, and analyst reports. The goal is to offer its wealth managers timely and convenient access to pertinent knowledge to enhance their ability to serve clients more effectively and efficiently. For example, the system can support a wealth manager in advising a specific client to revise an investment portfolio or develop a retirement plan.

Unlike earlier applications of AI and Machine Learning (ML), which were often specialized for specific tasks such as predicting customer churn or mastering the game of GO, emerging GenAI tools are general purpose, offer a high level of versatility, and can be employed in multiple tasks across various fields and domains (Eloundou et al., 2023). For example, the same GenAI tool could assist in drafting a go-to-market plan for a new product, developing computer code, creating varied versions of messages for a marketing campaign, or even generating multiple recipes based on a set list of ingredients. Thus, the power and versatility of GenAI have opened up new and exciting opportunities for enhancing knowledge management across diverse organizations and industries. This technology has launched a new era for knowledge management, ushering in unique opportunities (and challenges). These developments, in turn, offer future research trajectories for information systems researchers.

Electronic copy available at: https://ssrn.com/abstract=4782875

In this editorial, revisiting Alavi & Leidner (2001) as a conceptual lens, we consider the organizational implications of GenAI from a knowledge management perspective. While there are undoubtedly many implications of GenAI for organizations, a knowledge management perspective of GenAI can offer insights into some of the specific benefits and challenges of introducing GenAI into organizations. Knowledge management focuses on capturing, distributing, and effectively using knowledge. Generative AI impacts each of these processes, as it can generate novel insights, distribute codified knowledge efficiently, and assist in decision-making and problem-solving. These systems can transform the way knowledge is created, with the potential to shift the balance from mostly human-generated content to AI-generated insights. Moreover, understanding how AI interacts with existing knowledge structures, networks, and routines is pivotal to ascertain its impact on organizational learning, innovation, and competitiveness. By adopting a knowledge management lens, researchers can also assess the challenges and implications of AI, including potential knowledge silos, biases in AI-generated content, and the evolving role of humans in knowledge processes. In essence, this perspective provides a comprehensive framework to delve into both the transformative potentials and complexities introduced by generative AI in the organizational knowledge landscape.

The KM lens offers an insightful way to look at the effects of GenAI on organizations. However, there is a key distinction to be made. Traditional Knowledge Management Systems (KMS) categorize ”data” as basic facts, “information” as processed and contextualized data , and “knowledge” as insights drawn from human experience and judgement. In contrast, GenAI compresses the information layer and creates knowledge directly by processing very large volumes of data (Varun, 2023). While humans curate content in regular KMS, GenAI uses algorithms for this task. Therefore, the quality of GenAI’s output heavily relies on the data it is trained on.

Electronic copy available at: https://ssrn.com/abstract=4782875

Revisiting "what is knowledge"

To begin, we must revisit the question of "what is knowledge" in an organizational setting. The knowledge management literature defines knowledge as "a justified belief that increases an entity's capacity for effective action" (Alavi and Leidner, 2001, p. 109). An important part of this definition is that knowledge resides in an agent capable of taking action. Knowledge does not exist outside of an agent (a knower), and it is indelibly shaped by one's needs (the actions one needs to take) as well as one's initial stock of knowledge, which provides the basis for justifying the belief. GenAI does not necessarily alter the meaning of knowledge, but it does pose challenges to justifying the belief.

One may take for granted what GenAI produces, justifying the belief on the basis of GenAI having so much information at its disposal and being much "smarter" than a single human. Moreover, we must also consider the case of GenAI becoming a knowledge agent in itself- an agent capable of taking effective action (e.g., making decisions when embedded in workflows). This in turn-raises the question of how it justifies its beliefs in the knowledge it generates. Additionally, it leads us to consider the importance of regularly exposing the agent to new information and providing feedback to ensure that its actions remain effective and relevant. These challenges give rise to important questions. An assumption underpinning the knowledge-based view of the firm is that organizations can and should exercise oversight on the knowledge held by them and can and should facilitate knowledge processes, including the creation, storage, transfer, and application of organizational knowledge. By organizational knowledge, we refer to knowledge that resides in the members of the organization as well as organizational policies, procedures, and artifacts. The assumption is that organizations are the guardians of the knowledge within them and must put in place a system and culture that is oriented toward knowledge discovery, sharing, and use. This led

Electronic copy available at: https://ssrn.com/abstract=4782875

to systematic efforts to identify the knowledge, capture it, and store it in a well-managed system, to enable access (and hence, transfer) of the knowledge to others in the organization, and to ensure the appropriate application of knowledge across organizational units. Major goals of such knowledge management initiatives were to avoid reinventing the wheel, to learn from best practices, to know where/in whom knowledge resided in an organization and to appropriate valuable knowledge from one part of the firm to another. To foster such efforts, organizations built knowledge management systems that coordinated the organization's knowledge creation, storage, transfer, and application processes. As such, the organizations maintained control of what constituted knowledge in the system–what knowledge to include as well as who would be able to access what knowledge and when. Hence, the organization remained the central agent overseeing knowledge processes. The advent of GenAI portends a shift in the control of the knowledge processes in organizations with GenAI poised to potentially enhance and transform these processes – knowledge creation, knowledge storage, knowledge transfer, and knowledge application. We consider these in the following sections.

GenAI & Knowledge Creation

Organizations consistently generate, amplify, augment, and update both their tacit (unarticulated) and explicit (codified) knowledge resources through a variety of mechanisms and as knowledge flows through individuals, groups, and organizational artifacts (e.g., reports and databases). The prevalent view has been that knowledge is primarily created and expanded through social and collaborative processes among people as well as individual cognitive processes. Central to this viewpoint is the idea of individuals' agency in organizational knowledge creation. However, the individuals' capacity for creating knowledge is inherently constrained by the limitations of human

Electronic copy available at: https://ssrn.com/abstract=4782875

cognitive information processing, memory, and recall capability. This inherent limitation subsequently impacts the efficiency and pace at which new knowledge is generated and applied in organizations.

Four key modes of knowledge creation have been identified in the literature: socialization, externalization, combination, and internalization. These modes are interdependent; each mode draws upon and contributes to the other three modes, creating a dynamic and interconnected system for knowledge creation (for a description of the interactions among the modes, see Alavi & Leidner, 2001, pp. 116-117).

In our assessment, GenAI serves as a particularly useful tool in facilitating both the combination and internalization modes of knowledge creation. The combination mode involves synthesizing new knowledge by merging, categorizing, aggregating, and summarizing explicit knowledge from various sources. GenAI excels at generating new content, identifying patterns, and making predictions. It achieves this through navigation, analysis, and processing of massive volumes of data, obtained from a wide range of sources internal and external to the organization in a variety of formats (text, images, and audio). As a result, in the context of the combination mode of knowledge creation, GenAI systems expand the role and agency of IT. Rather than merely serving as a passive provider of a relatively limited set of pre-existing, codified knowledge captured by an organization, Gen-AI serves as an active knowledge co-creator by drawing on seemingly unlimited memory and information processing power for data analysis and prediction.

Some early use cases demonstrate this scenario. For instance in drug discovery, chemical language models (CLMs), a subset of large language models (LLMs), utilize text-based representations of chemical structures to identify potential drug molecules targeting specific disease-causing proteins. These models have, in some instances, surpassed traditional drug discovery methods.

Electronic copy available at: https://ssrn.com/abstract=4782875

Tools like Nvidia and AstraZeneca's MegaMolBART allow pharmaceutical companies to employ pre-trained CLMs for their projects (Moret, 2023). Beyond traditional small-molecule drugs, LLMs have also been instrumental in enhancing or designing new antibodies, which serve as therapies for conditions including cancers, and autoimmune disorders (Hie, 2023).

During the internalization process of knowledge creation, individuals develop new knowledge through cognitive processes such as learning and reflection. GenAI can facilitate and enrich these processes through interactive, conversational engagement between users and the GenAI analytical and predictive capabilities. GenAI can respond to user queries and provide coaching, contextual examples, and in-depth explanations. It can "scaffold" users through their tasks by suggesting actionable tips, outlining the next steps, and recommending best practices.

GenAI has the ability to "remember" the context and history of conversations with users. This enables it to deliver personalized responses and examples, which further enhance the learning experience and facilitate the retention of new knowledge. Organizations such as Khan Academy and DuoLingo have already integrated these and other GenAI functionalities into their learning support systems.

Within organizational settings, users can learn on the job through interactions with GenAI. For example, a computer programmer who regularly utilizes GenAI for coding tasks noted that GenAI occasionally "outperforms my own coding skills, and I learn from it." A field study focusing on customer support agents (Bynjolfsson, 2023) found that less-experienced agents who employed GenAI system performed their tasks more efficiently and learned faster compared to their peers who did not use the system and compared to those agents with more experience.

Thus far, we have noted the potential for GenAI to offer promising advancements in the knowledge-creation process, but its potential disruptions warrant attention. We see several areas

Electronic copy available at: https://ssrn.com/abstract=4782875

to note. First, in the past, organizations had the autonomy to decide which knowledge was vital and fostered a culture of sharing that knowledge. However, with GenAI tools like OpenAI's GPTs, which are trained on vast datasets, organizations often have limited influence over the foundational knowledge the models acquire. To tailor these tools to their specific needs, many organizations introduce their proprietary data for further training, a process known as domain adaptation or fine-tuning. While this offers some level of control, achieving complete oversight over GenAI's content, even with significant investments in fine-tuning, remains a challenge. In a sense, GenAI seems to be taking a more substantial role in organizational knowledge management.[2] In addition, even if organizations manage to exert full control over the content within their GenAI models, they might struggle to grant differentiated access levels to employees across various organizational levels and divisions. Second, by providing such readily available access to knowledge, employees might consult less with others for knowledge gathering. The socialization form of knowledge creation will become less important, and knowledge sharing, a behavior that has been encouraged for the past two decades, may fade. Knowledge networks in an organization might then shrink as individuals go first to the GenAI to seek knowledge and, given its seeming authority, reduce knowledge collaboration with others in the organization. But this then has subsequent consequences, such as a less homogenous organizational culture.

Third, the introduction of GenAI could potentially sidestep junior knowledge workers. Traditionally tasked with collecting information and drafting reports, these employees might now be overlooked as supervisors directly consult GenAI (Captain, 2023). This shift denies junior staff

[2] To realize benefit of GenAI, while mitigating its risks, organizations may use some techniques including Reinforcement Learning with Human Feedback (RLHF) (Griffin et al. 2013; Saunders et al.,2017) and tools that provide feedback to GenAI models (e.g., argilla.io) to exert more control over the knowledge generated by these tools. Organizations have a variety of techniques at their disposal to manage the content produced by GenAI. Nevertheless, achieving complete control over GenAI using these techniques remains challenging. Notably, preliminary research indicates the existence of "jailbreaking" methods that could potentially undermine these control techniques (Liu et al., 2023).

Electronic copy available at: https://ssrn.com/abstract=4782875

the chance to expand their knowledge and skills. While this may lead to fewer entry-level positions in the short term, it could also result in a future shortage of individuals adequately prepared for senior roles.

Fourth, in terms of internalization, whether an employee really internalizes the knowledge provided by GenAI or merely uses it in a rote fashion is open to question. Some employees may simply accept GenAI-generated knowledge as is and fail to internalize the knowledge for future use and development. This might have long-term negative consequences on employee learning and skills, including difficulties with onboarding as well as reskilling employees. Fifth, GenAI might dissuade employees from engaging in knowledge externalization – if employees regularly use the knowledge generated by GenAI to produce reports and plans but these new knowledge documents are not shared back with others or with the GenAI, then the very duplication of effort that previous knowledge management initiatives strove to eliminate might once again become common.

Lastly, GenAI models can occasionally generate outputs that are inaccurate, often referred to as the "hallucination" effect. Such inaccuracies could significantly impact an organization's knowledge-creation process. If employees don't thoroughly validate and cross-check the information from these models, false data could be injected into the organization's knowledge base. While there are techniques to curb these hallucinations, such as adjusting the model's settings for more deterministic outcomes (e.g., using a lower value for a hyperparameter referred to as *temperature*), organizations should establish formal procedures to ensure the authenticity of the content produced by GenAI models.

To summarize, GenAI tools can play a key and pro-active role in organizational knowledge creation by boosting individual learning and enhancing the combination mode of knowledge

Electronic copy available at: https://ssrn.com/abstract=4782875

generation. At the same time, GenAI might diminish the internalization, and externalization modes of knowledge creation. Some research questions follow from our discussion:

1- How does the integration of GenAI affect organizational culture, especially concerning knowledge sharing and collaboration among employees?
2- How does the pervasive use of GenAI affect the development and maintenance of social networks for knowledge creation within organizations?
3- How do employees judge the validity and relevance of knowledge?
4- To what extent does GenAI facilitate the internalization mode of knowledge creation, and are there potential risks of employees relying too much on GenAI output without deeply understanding the knowledge?
5- How might GenAI impact the externalization of knowledge, and what strategies can be employed to ensure that knowledge management doesn't regress due to duplicated efforts?

GenAI & Knowledge Storage and Retrieval

Knowledge in organizations is stored in a distributed fashion including individuals' minds, group and social systems, procedures, documents, and computer files. Tacit knowledge is typically stored within individuals, groups, and organizational culture, whereas explicit knowledge tends to be captured in formal and articulated forms in policies, procedures, and information systems. The ability to store organizational knowledge in the past was relatively limited. It was limited not only by the availability of data itself but also by the type of data. The introduction of knowledge management systems allowed organizations to store less structured data than previously, but an initial challenge facing organizations was getting employees to share their knowledge (i.e., store it in the knowledge management system). Employees were often reluctant to codify their tacit

Electronic copy available at: https://ssrn.com/abstract=4782875

knowledge and upload their explicit knowledge (e.g., reports, planning documents, client proposals, etc.) for fear of losing power and/or lack of time. Organizations sought cultural as well as incentive solutions to encourage employees to contribute (store) their knowledge to (in) organizational knowledge management systems. As employees became more willing to share their knowledge, in part because of incentives provided by their organization, the next challenge became distinguishing between highly valuable knowledge and less valuable knowledge – as long as employees were rewarded for the act of sharing regardless of the quality of knowledge shared, the knowledge systems would expand but the usefulness remain the same or even decline as the time to find useful knowledge would increase with the increased knowledge available. Organizations then derived feedback mechanisms – rating systems – whereby the knowledge shared by employees could be rated by others. Many companies found that knowledge maps – maps of who knew what – were more effective than storing knowledge itself because the maps required less upkeep than the vast knowledge stores that organizations were managing. Yet knowledge maps introduced their own challenges – such as bias in the assessment of one's own expertise.

GenAI alleviates some of the previous challenges of knowledge storage while creating new ones. GenAI introduces new sources of potential knowledge -- e.g., knowledge captured in encyclopedias, dictionaries, books, scientific papers, literature, press etc. – that does not require nudging to obtain. And GenAI itself produces new knowledge, or in some cases existing knowledge newly synthesized – summary and integration of reports and documents, responses to queries, and capturing meeting notes, to name a few. The knowledge is then stored and made accessible for future reference.

Among the goals of storing vast amounts of organizational knowledge has been to provide employees with high-quality knowledge when they need it in an easily accessible manner. It has

Electronic copy available at: https://ssrn.com/abstract=4782875

long been observed that when given the choice between information that is readily available but not of high quality and information that is of high quality but hard to access, individuals will choose the readily available information (Simon, 1956). People satisfice when it comes to knowledge-seeking behavior. The quantity and speed of knowledge accessibility provided by GenAI is heretofore unparalleled. There is little doubt that knowledge accessed through GenAI will be widely used (and in some cases, perhaps misused). The question of quality is still open to debate. Research questions to consider:

1- How does the integration of GenAI impact the intrinsic and extrinsic motivation of employees to share or codify their tacit knowledge?
2- How can traditional knowledge vetting and rating systems (e.g., Davenport and Prusak, 1998) be repurposed to evaluate content stored and retrieved by GenAI to ensure quality?
3- How can organizations integrate human-derived tacit knowledge and GenAI-generated explicit knowledge to optimize knowledge management systems?
4- How can organizations use GenAI tools to reliably and independently rate and evaluate employee-generated knowledge?
5- How can organizations use employees' queries to GenAI tools (if controlled by the organization) to automatically capture and document employees' knowledge?

GenAI & Knowledge Transfer

Organizations do not have a central memory that encompasses and can recall all their knowledge. This makes it very challenging for organizations to know what they know, and where the knowledge resides, and who needs it when. Thus, a key process is transferring knowledge from its source (where it resides) to where it is needed in a timely fashion. Information systems in form of databases and knowledge directories have been used to enhance this process. Nevertheless, a

Electronic copy available at: https://ssrn.com/abstract=4782875

number of factors had made this a difficult and cumbersome. These factors include the type of knowledge (it is difficult to transfer tacit knowledge), the complexity of navigating and querying databases, the difficulty of locating the required knowledge, and organizational reward systems that often encourage knowledge hoarding.

GenAI tools can support and improve knowledge transfer in organizations. For example, GenAI can facilitate knowledge transfer during the onboarding of new employees. By analyzing a new employee's initial responses and role, the AI can tailor content to be both relevant and engaging, providing an easier integration into the organization. This includes interactive tutorials and simulations, particularly beneficial for roles requiring hands-on learning and experience.

Another advantage to knowledge transfer is the instantaneous nature of AI. Users can receive immediate answers to their questions, a boon for globally distributed teams where time zones can delay communication. GenAI may also reduce communication barriers in knowledge transfer because employees may be less inhibited when it comes to asking GenAI a question than asking their supervisor or coworkers. This ensures that consistent and accurate information is transferred and communicated.

While GenAI offers many benefits, it can pose some new challenges to knowledge transfer. Depending on how the AI model is initially developed, its knowledge can come from multiple sources within and outside the organization. Hence, the organization may lose control of what knowledge is transferred to employees. Thus, whereas in the past, a major challenge of knowledge management was ensuring the maximum transfer of knowledge, now, with the introduction of GenAI, the challenge has shifted. Organizations must be cautious about the information GenAI shares with employees. Some of this knowledge may contain privileged information that shouldn't be disseminated. Moreover, an overabundance of information provided by GenAI may adversely

Electronic copy available at: https://ssrn.com/abstract=4782875

impact an employees' intrinsic knowledge-seeking tendencies and potentially curtail their creative capacities. Finally, whereas previous knowledge management increased transfer by extending an individual's reach beyond the formal communication channels, thereby increasing an individual's knowledge of the expertise held by other employees and business units and extending their internal social capital, GenAI might result in employees learning the knowledge but not learning as much about other employees and business units. Hence, to preserve team cohesion and facilitate cultural assimilation within organizations, it is imperative to establish an equilibrium between employees' utilization of GenAI and their interpersonal interactions, as the latter can inherently promote unity and cultural integration.

Given the far-reaching implications of GenAI on knowledge transfer, researchers can study and answer a variety of questions related to this topic. A few questions we find worth answering include:

1- To what extent organizations can/ should use internal vs. external data to train GenAI and how might the balance of internal and external data sources impact the accuracy and relevance of GenAI outputs in knowledge transfer scenarios?

2- How does GenAI influence the overall rate and level of knowledge sharing in various industries?

3- To what extent does GenAI's content curation improve the efficiency and relevance of knowledge transfer in organizations?

4- What are the potential risks of over-reliance on GenAI for knowledge transfer with respect to disseminating sensitive and privileged information?

Electronic copy available at: https://ssrn.com/abstract=4782875

5- To what extent does the excessive dependence on GenAI during employee onboarding or training hinder the effective assimilation of employees into the established organizational culture?

GenAI & Knowledge Application

Knowledge Application is a key element of knowledge management in organizations. Storage and availability of knowledge do not necessarily lead to improved productivity and performance. Value is created only when knowledge is applied to take effective action. Organizational knowledge is applied through various mechanisms including enforcing policies and procedures and establishing workflows and best practices.

GenAI can increase the speed, consistency, and level of organizational knowledge applications by facilitating search and access to stored knowledge through natural language interactions with users and responding to their context-specific questions and instructions. Case observations and research studies that are emerging are showing positive productivity and quality impact of knowledge application enhanced by GenAI systems on knowledge work. For example, Brynjolfsson et. al (2023) showed that capturing, combining, and applying customer support knowledge via GenAI capabilities led to a 14% increase in the overall productivity of customer support agents, with a 30% increase in the productivity of novice agents. Furthermore, overall, the quality of customer service was improved as measured and reflected in customer satisfaction scores.

A few initial use cases have showcased the potential of GenAI tools in applying knowledge and enhancing performance. For instance, Intercom, a company focused on customer service solutions, has initiated pilot programs that leverage GenAI to enhance customer engagement and drive corporate growth. In another instance, J.P. Morgan is using GenAI to generate synthetic data to

Electronic copy available at: https://ssrn.com/abstract=4782875

train machine learning models that detect new and emerging anomalies that they might not be able to detect if they were only trained on real data (JP Morgan, 2023).

Moreover, there is a trend to integrate GenAI capabilities into enterprise software systems. As this trend continues, knowledge management processes can become increasingly efficient when embedded in various workflows across the organization, time zones, and locations. For example, meetings conducted via Microsoft Teams can be automatically transcribed and summarized, with suggestions for follow-up meeting times and draft agendas, thus streamlining notetaking, scheduling and agenda development tasks. Another example is GenAI capabilities embedded in word processing software that can be applied to drafting and/or editing written reports.

Applying GenAI to knowledge management processes in organizations is expected to increase productivity and quality of knowledge work across all industries. Goldman Sachs (2023) estimates that GenAI alone could raise annual US labor-productivity growth by just under 1.5 percentage points, following widespread adoption. In a controlled experiment, software developers who used GitHub Copilot GenAI completed their task (implementing an HTTP server in JavaScript) 55.8% faster than those who did not have access to GitHub Copilot (Peng et al, 2023). In another experiment, participants (college-educated professionals) who used GenAI (ChatGPT, in this case) in completing a writing task were 40% more productive compared to participants who did not use GenAI. Furthermore, the quality of output of those who used ChatGPT rose by 18% (Noy and Zhang, 2023).

While GenAI has the potential to significantly enhance productivity, realizing these benefits hinges on employees' willingness to effectively leveraging these tools. If employees perceive GenAI as a means to amplify their knowledge or task capabilities, they are more likely to recognize its value, resulting in increased usage and task productivity. Conversely, if they view GenAI as a threat,

Electronic copy available at: https://ssrn.com/abstract=4782875

believing it can produce superior results compared to their own abilities, they may be discouraged from using the system, thinking that it diminishes their value to the organization. Consequently, the acceptance or resistance towards GenAI as a knowledge system can vary, ultimately affecting the overall productivity of GenAI tools. Furthermore, there is also the possibility of productivity losses when it comes to using GenAI. due to inadequate training. First, while the potential for the wrong and/or inaccurate application of knowledge existed with knowledge management systems in the past, this risk was reduced by the careful and intentional knowledge curation. Without such intentionality, the knowledge contained in GenAI might be misapplied if employees rush to apply the knowledge without taking the time to carefully evaluate it. The time saved from an extensive search is partially taken by the need for carefully evaluating the veracity and applicability of the knowledge. In addition, employees may spend time asking irrelevant questions of GenAI, much as employees spent time surfing the web during work hours when the internet first appeared, reducing the productivity gains of the internet for their personal work. Second, applying the knowledge generated by GenAI tools may yield intellectual property risks that can take two forms: One is leaking the firm's intellectual, confidential, and protected information by including it in GenAI prompts and queries. The second is unauthorized access and use of information protected by copyright, intellectual property, and privacy laws, which may in turn lead to legal challenges. For instance, Getty Images, a photography agency, sued Stability AI (a GenAI model vendor) for using Getty's copyrighted material without permission (Brittain, 2023).

Third, large organizations that possess extensive data, advanced computational resources, and a skilled talent pool appear to be better poised to harness the benefits of GenAI compared to their smaller counterparts with limited resources. This differential access and capability could amplify the competitive edge of these larger entities. Consequently, smaller organizations might find

Electronic copy available at: https://ssrn.com/abstract=4782875

themselves at a competitive disadvantage, potentially widening the gap between the two in terms of technological prowess and market dominance, which may in turn impact consumer welfare. There are several questions that arise regarding the application of knowledge generated by GenAI tools:

1. What policies and safeguard need to be developed to guide effective application of GenAI in organizational knowledge management while mitigating their risks?
2. To what extent does GenAI-enhanced knowledge management impact the overall productivity and innovation of an organization?
3. What are the best approaches to training knowledge workers to effectively apply GenAI in support of their activities?
4. How should knowledge work and knowledge flows be re-designed to harness GenAI capabilities?
5. Which domains and use cases benefit from the integration of GenAI, and in which areas might its use be counterproductive?

Conclusion

The focus on IT's contribution to knowledge management (Alavi and Leidner,2001; Alavi and Denford, 2011) paved the way for research and exploration in this domain. The subsequent work not only provided insights into the nuances of knowledge management but also highlighted the intricacies of IT's integration within this domain.

When we shift our focus to GenAI and its prospective influence on organizational knowledge management processes, it is apparent that we are standing at the precipice of a transformative period. Building on the foundational works of predecessors and the burgeoning advancements in

Electronic copy available at: https://ssrn.com/abstract=4782875

GenAI, our prediction is anchored in a realistic yet optimistic vision. In the near future, we anticipate that GenAI will play a tangibly significant role as an integral component, in the knowledge management processes we previously delineated. While its full potential remains to be explored, the initial indicators hint at GenAI becoming a force in reshaping the organizational knowledge management landscape. Over the longer term, we envision GenAI further shifting the role of information technology from facility knowledge processes to providing wisdom. Today, information technology predominantly aids in facilitating knowledge management processes. It helps gather, store, and disseminate information. However, with the advent and continuous refinement of GenAI, we foresee a transformative shift in this paradigm. Rather than just enabling processes, GenAI will likely play a more profound role. It could sift through vast troves of data to identify patterns, offer insights, and predict outcomes, effectively providing organizations with wisdom rather than just information. This "wisdom" translates to actionable insights, strategic foresight, and nuanced understanding that can be applied across various organizational functions to drive innovation, efficiency, and growth. Thus, in the long run, GenAI's impact is projected to be transformative, pushing the boundaries of information technology's capabilities within organizational knowledge management.

Finally, in this editorial, we highlighted the significant impact of GenAI on organizational knowledge management, emphasizing how GenAI can influence each of the four knowledge management processes. As we alluded to in our discussions, organizations have compelling reasons for transforming their knowledge management practices by leveraging GenAI. In doing so, organizations must assess and adapt their knowledge management practices to incorporate GenAI effectively. While traditional KMS allow for inputs to be carefully reviewed and feedback mechanisms to clarify the knowledge's origins, context, and contributors, GenAI operates

Electronic copy available at: https://ssrn.com/abstract=4782875

differently. GenAI functions as a black box, making it challenging to fully comprehend the reasoning behind its recommendations. The more GenAI takes over decision-making and processes, the less active involvement and cognitive engagement employees have, and the greater the reliance on algorithmic trust. Therefore, it is important to fully understand the long-term effects of GenAI on organizational KM (perhaps through small-scale pilot~~s~~ use cases within the organization) to better tailor GenAI's capabilities to meet organizational needs. This cautious approach ensures a smoother transition to GenAI while maintaining transparency and alignment with organizational requirements. Given these intricacies in adopting GenAI in organizational KM processes, this integration presents a plethora of research opportunities for scholars. Table 1 provides a summary of GenAI's potential effects on organizational knowledge management and the corresponding research inquiries.

Table 1. Knowledge Management Processes and the Potential Role of GenAI

KM Processes	Potential Role of GenAI	Research Questions
Creation	• GenAI is adept at synthesizing new knowledge by merging, categorizing, aggregating, and summarizing explicit knowledge from varied sources. • GenAI aids in the cognitive processes of individuals, like learning and reflection, by offering coaching, contextual examples, in-depth explanations, and actionable recommendations. • GenAI can process tacit knowledge (e.g., employees' prompts and meeting notes) to create explicit knowledge. • GenAI can improve the productivity of knowledge workers. • Less-experienced knowledge workers can use GenAI to faster and perform tasks more efficiently.	1. How does the integration of GenAI affect organizational culture, especially concerning knowledge sharing and collaboration among employees? 2. How does the pervasive use of GenAI affect the development and maintenance of social networks for knowledge creation within organizations? 3. How do employees judge the validity and relevance of knowledge? 4. To what extent does GenAI facilitate the internalization process of knowledge creation, and are there potential risks of employees relying too much on GenAI output without deeply understanding the knowledge? 5. How might GenAI impact the externalization of knowledge, and what strategies can be employed to ensure that knowledge management doesn't regress due to duplicated efforts?

Electronic copy available at: https://ssrn.com/abstract=4782875

Storage and Retrieval	• GenAI can retrieve and store knowledge from different sources, including individual minds (e.g., employees' prompts), group dynamics (e.g., meeting notes), documents, and computer files. • The knowledge made available by GenAI is always readily and instantly available. • GenAI enhances traditional Knowledge Management Systems (KMS) by optimizing the storage and retrieval of unstructured data, such as text. • GenAI may democratize access to knowledge within organizations. • GenAI can be used to sort and rank the collective knowledge within organizations.	1. How does the integration of GenAI impact the intrinsic and extrinsic motivation of employees to share or codify their tacit knowledge? 2. Can traditional knowledge vetting and rating systems (e.g., Davenport and Prusak, 1998) be repurposed to evaluate content stored and retrieved by GenAI to ensure quality? 3. How can organizations integrate human-derived tacit knowledge and GenAI-generated explicit knowledge to optimize knowledge management systems? 4. How can organizations use GenAI tools to reliably and independently rate and evaluate employee-generated knowledge? 5. How can organizations use employees' queries to GenAI tools (if controlled by the organization) to automatically capture and document employees' implicit knowledge?
Transfer	• By eliminating the hurdles in traditional KMS (e.g., difficulty in transferring tacit knowledge, complex navigation and querying, and challenges in locating necessary knowledge), GenAI can easily facilitate knowledge transfer within organizations. • GenAI can be used to create customized/ interactive onboarding programs to facilitate knowledge transfer to new employees. • GenAI can provide interactive tutorials and simulations, which are especially valuable for roles demanding practical learning and experience. • GenAI can foster a learning culture by eliminating employees' potential reluctance to ask questions from their superiors.	1. To what extent organizations can/ should use internal vs. external data to train GenAI and how might the balance of internal and external data sources impact the accuracy and relevance of GenAI outputs in knowledge transfer scenarios? 2. How does GenAI influence the overall rate and level of knowledge sharing in various industries? 3. To what extent does GenAI's content curation improve the efficiency and relevance of knowledge transfer in organizations? 4. What are the potential risks of over-reliance on GenAI for knowledge transfer with respect to disseminating sensitive and privileged information? 5. To what extent does the excessive dependence on GenAI during employee onboarding or training hinder the effective assimilation of employees into the established organizational culture?
Application	• GenAI enhances speed, consistency, and efficiency in applying organizational knowledge. • GenAI aids in searching and accessing stored knowledge through natural language interactions, responding to context-specific user queries. • GenAI can streamline the use of organizational knowledge to address business challenges, such as using ChatBots to respond to customer inquiries based on company knowledge. • GenAI can tailor knowledge based on specific employee preferences, needs,	1. What policies and safeguard need to be developed to guide effective application of GenAI in organizational knowledge management while mitigating their risks? 2. To what extent does GenAI-enhanced knowledge management impact the overall productivity and innovation of an organization?? 3. What are the best approaches to training knowledge workers to effectively apply GenAI in support of their activities? 4. How should knowledge work and knowledge flows be re-designed to harness GenAI capabilities?

Electronic copy available at: https://ssrn.com/abstract=4782875

	or historical interactions. This can enhance knowledge internalization and, subsequently, knowledge application. • GenAI can analyze complex challenges by accessing varied knowledge sources and establishing relevant connections. • GenAI can democratize specialized or niche knowledge, making it available to a broader audience and ensuring a more level playing field.	5. Which domains and use cases benefit from the integration of GenAI, and in which areas might its use be counterproductive?

References

Alavi, M. Denford, James, (2011). Knowledge Management: Process, Practice, and Web 2.0, Handbook of Organizational Learning and Knowledge Management, Easterby-Smith, M., and Lyles, M. Edits, John Wiley and Sons, Inc.

Alavi, M. and Leidner, D., E. (2001). Knowledge Management and Knowledge Management Systems: Conceptual Foundations and Research Issues." MIS Quarterly, 25(1), PP. 107-136.

Brittain, B. (2023, February 6). Getty Images lawsuit says Stability AI misused photos to train AI. Reuters. https://www.reuters.com/legal/getty-images-lawsuit-says-stability-ai-misused-photos-train-ai-2023-02-06/

Brynjolfsson, E., Li, D., & Raymond, L. R. (2023). Generative AI at work (No. w31161). National Bureau of Economic Research.

Davenport, T. H., & Prusak, L. (1998). Working knowledge: How organizations manage what they know. Harvard Business Press.

Eloundou T, Manning S, Mishkin P, Rock D. (2023). GPTs are GPTs: An early look at the labor market impact potential of large language models, arXiv preprint arXiv:2303. 10130.

Goldman Sachs, Generative AI could raise global GDP by 7%. (2023, April 5). Goldman Sachs. https://www.goldmansachs.com/intelligence/pages/generative-ai-could-raise-global-gdp-by-7-percent.html

Griffith, S., Subramanian, K., Scholz, J., Isbell, C. L., & Thomaz, A. L. (2013). Policy Shaping: Integrating Human Feedback with Reinforcement Learning. Neural Information Processing Systems, 26, 2625–2633. https://papers.nips.cc/paper/5187-policy-shaping-integrating-human-feedback-with-reinforcement-learning.pdf

Grover, V. (2023), Personal Communications.

Electronic copy available at: https://ssrn.com/abstract=4782875

Hie, B., Shanker, V. R., Xu, D., Bruun, T., Weidenbacher, P. A., Tang, S., Wu, W., Pak, J. E., & Kim, P. S. (2023). Efficient evolution of human antibodies from general protein language models. Nature Biotechnology. https://doi.org/10.1038/s41587-023-01763-2

Liu, Y., Deng, G., Xu, Z., Li, Y., Zheng, Y., Zhang, Y., ... & Liu, Y. (2023). Jailbreaking ChatGPT via prompt engineering: An empirical study. arXiv preprint arXiv:2305.13860.

J.P. Morgan, The AI revolution for payments & tech | J.P. Morgan. (2023). https://www.jpmorgan.com/payments/payments-unbound/volume-3/smart-money

McKinsey & Company, Meet Lilli, our generative AI tool that's a researcher, a time saver, and an inspiration. (2023, August 16). McKinsey & Company. https://www.mckinsey.com/about-us/new-at-mckinsey-blog/meet-lilli-our-generative-ai-tool

Meta Platforms, Inc. [Meta]. (2023). Llama 2: open source, free for research and commercial use. ai.meta.com. https://ai.meta.com/resources/models-and-libraries/llama/

Moret, M., Pachón-Angona, I., Cotos, L., Yan, S., Atz, K., Brunner, C., Baumgärtner, M., Grisoni, F., & Schneider, G. (2023). Leveraging molecular structure and bioactivity with chemical language models for de novo drug design. Nature Communications, 14(1). https://doi.org/10.1038/s41467-022-35692-6

Noy, S., Zhang, W. (2023) Experimental evidence on the productivity effects of generative artificial intelligence, Science, 381(6654), pp. 187-192.

Peng S, Kalliamvakou E, Cihon P, Demirer M. (2023) The impact of AI on developer productivity: Evidence from GitHub copilot. arXiv preprint arXiv:2302.06590.

Salesforce. (2023). Top Generative AI Statistics for 2023. Salesforce. https://www.salesforce.com/news/stories/generative-ai-statistics/#:~:text=Salesforce%27s%20latest%20State%20of%20IT,is%20a%20%E2%80%9Cgame%20changer.%E2%80%9D

Saunders, W. S., Sastry, G., Stuhlmüller, A., & Evans, O. (2018). Trial without Error: Towards Safe Reinforcement Learning via Human Intervention. arXiv (Cornell University), 2067–2069. https://arxiv.org/abs/1707.05173

Simon, H. A. (1956). Rational choice and the structure of the environment. Psychological review, 63(2), 129.

Sinha, S. (2023). From GPT-1 to GPT-4: A look at the evolution of Generative AI. HGS. https://hgs.cx/blog/from-gpt-1-to-gpt-4-a-look-at-the-evolution-of-generative-ai/#:~:text=Like%20previous%20models%2C%20GPT%2D4,technique%20into%20its%20training%20process.

Electronic copy available at: https://ssrn.com/abstract=4782875

Son, H. (2023, March 15). Morgan Stanley is testing an OpenAI-powered chatbot for its 16,000 financial advisors. CNBC. https://www.cnbc.com/2023/03/14/morgan-stanley-testing-openai-powered-chatbot-for-its-financial-advisors.html

Spisak, B., Rosenberg, L.B., and Beilby, M. (2023). 13 Principles for Using AI Responsibly. HBR. https://hbr.org/2023/06/13-principles-for-using-ai-responsibly

Author Bios

Maryam Alavi (Maryam.alavi@Scheller.gatech.edu) is a professor of IT Management and holds the Elizabeth T. and Thomas M. Holder Chair at the Scheller College of Business at Georgia Institute of Technology. She served as dean of the College from 2014-2023. Dr. Alavi is a Fellow of the Association for Information Systems and was awarded the distinguished Marvin Bower Faculty Fellowship at Harvard Business School. She received her PhD from the Ohio State University, and her current research focuses on digital innovation and AI support of knowledge management and knowledge work.

Dorothy E. Leidner (dorothy@virginia.edu) is the Leslie H. Goldberg Jefferson Scholars Foundation Distinguished Professor of Business Ethics at the University of Virginia. She is a LEO and a Fellow of the Association of Information Systems. Dorothy received her PhD in information systems from the University of Texas at Austin and holds an honorary doctorate from Lund University. She is a professional research fellow with Deakin University in Australia and a visiting professor with the University of Gothenburg, Sweden. Her current research focuses on the ethics of personal data digitalization.

Reza Mousavi (mousavi@virginia.edu) is an assistant professor in the McIntire School of Commerce at the University of Virginia. Reza received his Ph.D. in Business Administration from Arizona State University. His research emphasizes on the societal impacts and economics of social media, natural language processing (NLP), artificial intelligence and business analytics, and healthcare information systems. Before joining academia, Reza spearheaded numerous AI and data science initiatives for Fortune 50 corporations and premier consulting firms.

Electronic copy available at: https://ssrn.com/abstract=4782875